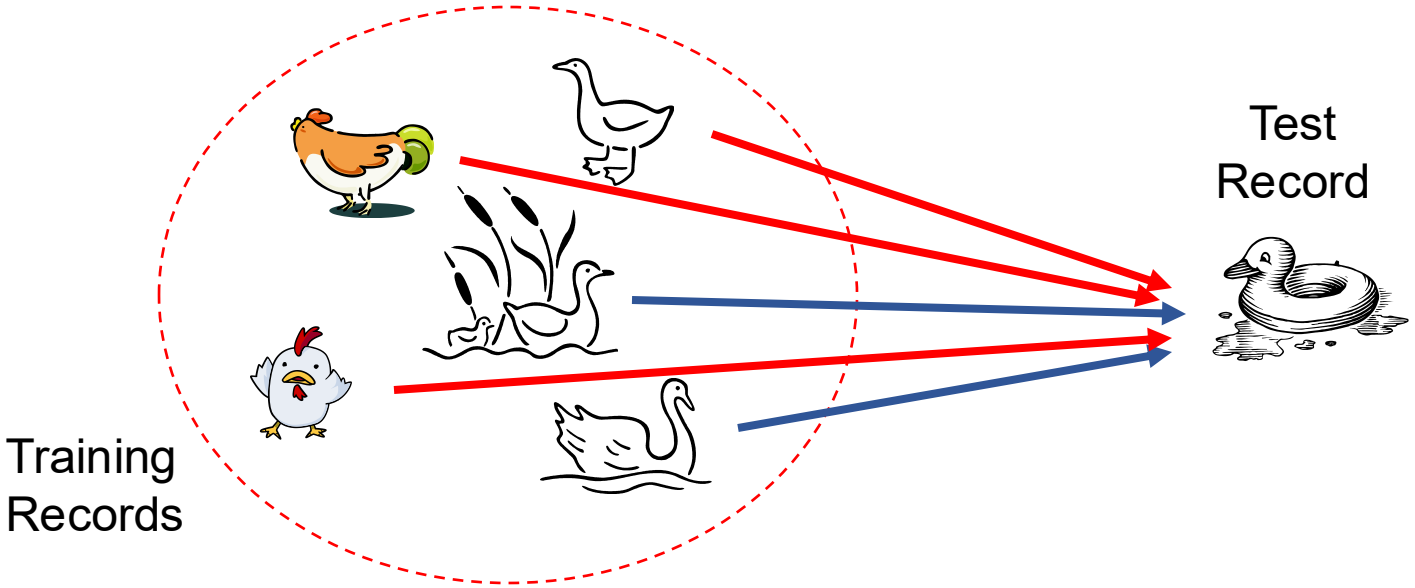
ITEC 4220 Adv Data Analytics (Modified from CIS4930: Data Science Foundations)

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K Nearest Neighbors (KNN)

KNN: Basic Idea

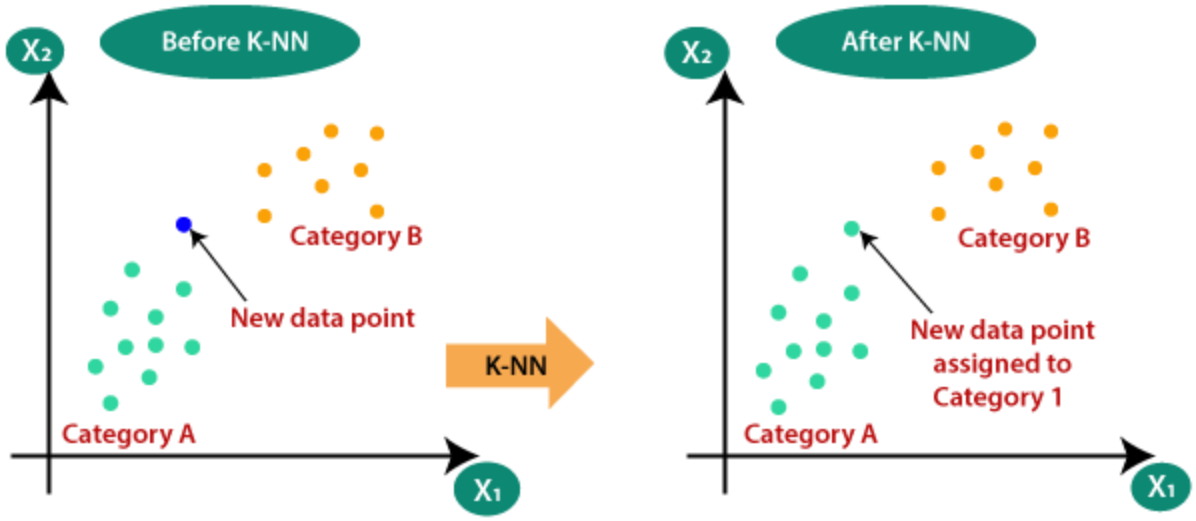
- Simple idea: classify a record like similar records.
- If it walks like a duck, quacks like a duck, then it's probably a duck.



KNN: Basic Idea

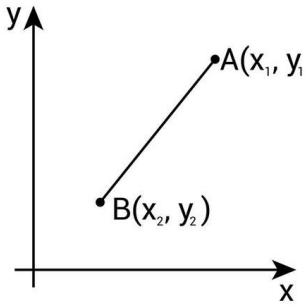
- Relies on the idea that similar records exist near each other.
- For a given record to be classified, identify nearby records.
  - ✓"Near" means records with similar predictor values $X_1, X_2, \ldots X_p$
- Classify the record based on the class of the "neighbors".
- KNN works well with:
  - ✓Smaller datasets
  - ✓Few features

KNN: Basic Idea



How to Measure “Nearby”?

- The most popular distance measure is **Euclidean distance**



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Data Scaling

- k-NN is distance-based: It calculates distances (e.g., Euclidean) between data points.
- Features with larger scales dominate the distance metric, leading to biased results.
- Example:
  - ✓ Feature A (Age): 20–60
  - ✓ Feature B (Income): 30,000–120,000
  - ✓ Income will disproportionately affect the distance
- We need to to **scale** data to bring all of the variables within the same scale.

Data Scaling

- Scaling means adjusting the range of features, so they are on a similar scale.
- Example:
  - ✓Age: 0–100
  - ✓Income: 20,000–200,000
- Without scaling, income dominates distance (in KNN) or activation function calculation (in NN) simply because of its bigger numbers.

Data Scaling

- Common methods:
  - ✓Standardization (StandardScaler): make features to have mean = 0 and standard deviation = 1.
    - Use when features have different averages and standard deviation.
  - ✓Normalization (MinMaxScaler): rescale features to [0,1].
    - Use when you want features strictly in a range (like 0–1 for images).

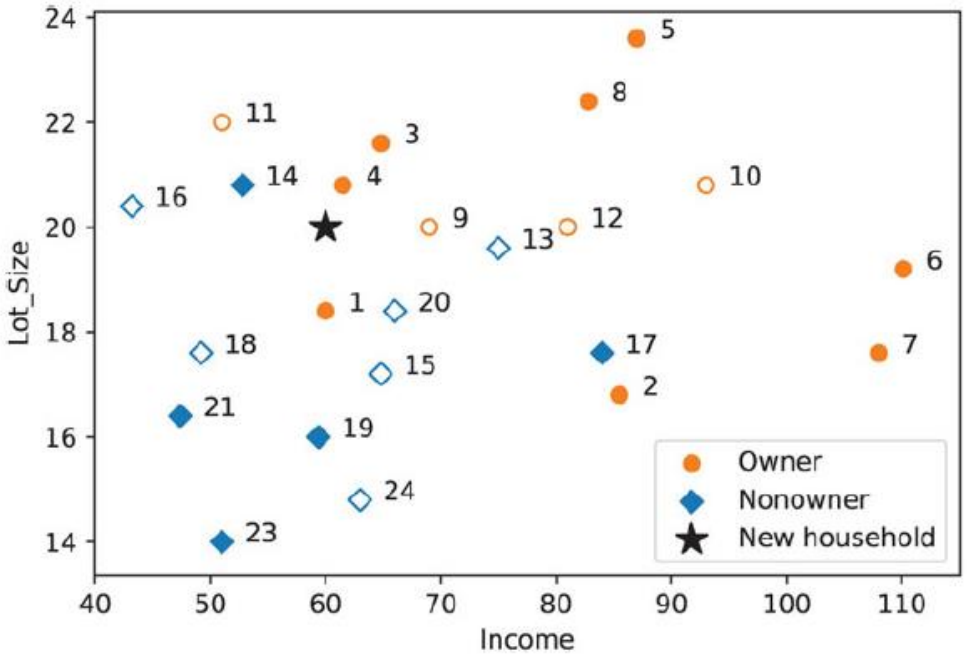
How Many Neighbors?

- *K* is the number of nearby neighbors to be used to classify the new record.
  - ✓*K*=1 means use the single nearest record.
  - ✓*K*=5 means use the 5 nearest records.
- Typically choose that value of k which has lowest error rate in validation data.

Example

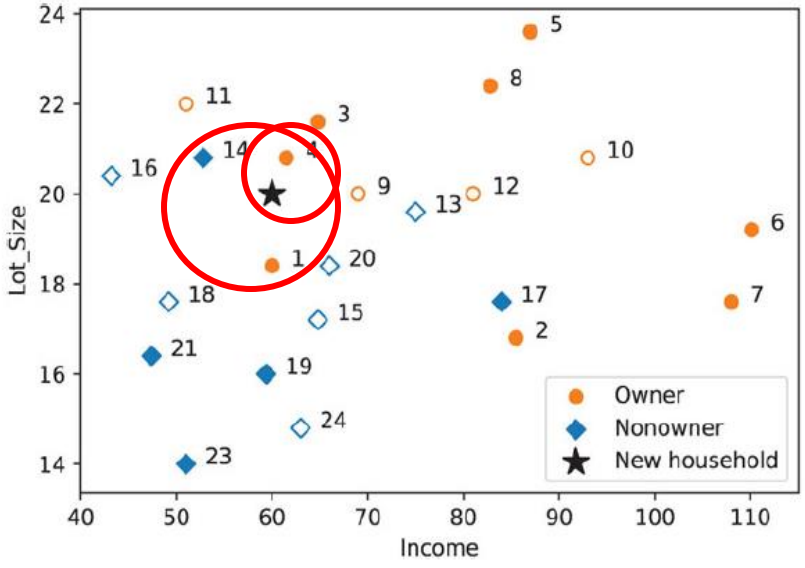
- A riding mower company wants to classify families to those likely to purchase a mower and not purchasing.

Household number	Income ($000s)	Lot size (000s ft^2)	Ownership of riding mower
1	60.0	18.4	Owner
2	85.5	16.8	Owner
3	64.8	21.6	Owner
4	61.5	20.8	Owner
5	87.0	23.6	Owner
6	110.1	19.2	Owner
7	108.0	17.6	Owner
8	82.8	22.4	Owner
9	69.0	20.0	Owner
10	93.0	20.8	Owner
11	51.0	22.0	Owner
12	81.0	20.0	Owner
13	75.0	19.6	Nonowner
14	52.8	20.8	Nonowner
15	64.8	17.2	Nonowner
16	43.2	20.4	Nonowner
17	84.0	17.6	Nonowner
18	49.2	17.6	Nonowner
19	59.4	16.0	Nonowner
20	66.0	18.4	Nonowner
21	47.4	16.4	Nonowner
22	33.0	18.8	Nonowner
23	51.0	14.0	Nonowner
24	63.0	14.8	Nonowner
25	60.0	20.0	?



Example

- If k=1, the closest neighbor is #4
  - ✓ The closest neighbor is owner, so the new household is classified as owner
- If k=3, the three closest neighbors are #4, #14, & #1
  - ✓ The majority vote is owner, so the new household is classified as owner




ML
K NEAREST
NEIGHBORS